Core-First Language Theory (CFLT): A Discourse-Level Linearization Protocol for Cross-Linguistic Communication and LLM Prompting

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Abstract

Second language acquisition (SLA) and Large Language Model (LLM) prompting share a structural bottleneck: the cost of linearizing a multidimensional semantic representation into a single ordered string. Adult second-language (L2) learners pay it as working-memory overhead; LLMs pay it as order-sensitivity and the "Lost in the Middle"positional-accuracy effect (Liu et al. 2023). This paper presents Core-First Language Theory (CFLT), a normative discourse-level linearization protocol that fixes the relative order of two constituent tiers as `[Core] → [Reason] → [Space] → [Time]`: the speaker's salience anchor (the event, identity, state, or request being committed to) occupies sequence-initial position —not a verb, and not a value judgment about "the most important word"—and the circumstantial ground frame of reason, place, and time follows. CFLT operates above the morphosyntactic level: each language assembles the event nucleus (the Core) in its native syntax, while the protocol governs only the order of the ground frame and the boundary between the two. Fixing the slot order moves the linearization decision out of the producer's real-time workload: the L2 learner substitutes lexical material into a stable scaffold rather than restructuring a plan shaped by their first language (L1), and the model is intended to meet the task in its high-attention prefix (a mechanism the paper predicts but does not yet measure). A pilot study (24 cases × 2 conditions × 3 runs × 5 frontier models = 720 trials) provides preliminary evidence that Core-first prompts raise extraction accuracy on the distractor-heavy condition (Level 3) from a mean of 65.6% to 100% across all five models and reduce completion tokens by up to 38% on reasoning models with visible chain-of-thought, with no effect on short-output models; on a separate buried-decision condition (Level 4) one model (DeepSeek V4 Pro) shows no benefit (−11 pp) while the other four saturate. We frame these results as suggestive, not confirmatory, and position CFLT as a language-neutral interface for human–AI reasoning whose falsification conditions are set out as an empirical agenda. Reproducible data, prompts, and evaluation scripts are available at https://github.com/corefirst/cflt.

Keywords: linearization, salience anchor, second language acquisition, prompt engineering, prompt order-sensitivity, Transformer primacy, cognitive ergonomics, discourse protocol

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1. Introduction

Core-First Language Theory (CFLT)[1] is a discourse-level protocol that fixes the relative order of two constituent tiers —`[Core] → [Reason] → [Space] → [Time]` —while leaving each language's internal morphosyntax untouched. The Core is the speaker's *salience anchor* (the event, identity, state, or request being committed to), placed first; the ground frame of reason, place, and time follows. Concretely, it rewrites "*Yesterday, at the cafe across from the station, because the wifi was unstable, please summarize the attached document into three bullet points*" as "*Please summarize the attached document into three bullet points, because the wifi was unstable, at the cafe across from the station, yesterday*"—same words, Core moved to the front.

CFLT targets a cost that adult second-language (L2) production and LLM prompting both incur: restructuring a multi-dimensional semantic representation into a single ordered string. Learners pay it as working-memory overhead; LLMs pay it as order-sensitivity and the "Lost in the Middle"positional-accuracy effect (Liu et al. 2023).

The mechanism is to move the linearization decision out of the producer's real-time workload: with the slot order fixed, the learner substitutes words into a stable scaffold instead of restructuring an L1-shaped plan into L2 syntax, and the model is intended to meet the task in its high-attention prefix rather than search for it mid-sequence —the same intervention working through partially distinct mechanisms on the two sides (the LLM-side mechanism is predicted in §3.2, not yet directly measured). In a pilot across five frontier models, Core-first reordering raised accuracy on the distractor-heavy condition (one of four difficulty levels) from a mean of 65.6% to 100% and cut completion tokens by up to 38% on the two reasoning models with visible traces, though one model showed no benefit on a buried-decision variant —suggestive, not confirmatory, evidence (§6).

[1] Not to be confused with Ambridge & Wagner (eds. 2021), *Testable Theories of Core First Language Acquisition* (*Journal of Child Language*, Special Issue 48/S5): that work parses as `[core] + [first-language-acquisition]` and concerns how children acquire their L1, whereas CFLT parses as `[core-first] + [language-theory]` and concerns the core-first sequencing rule for cross-linguistic discourse. We flag the distinction because retrieval systems surface both together.

The gap between thought and speech is bridged by linearization: the mapping of a multidimensional semantic representation onto a sequential string of words. Natural languages adopt different default linearizations —SVO versus SOV, head-initial versus head-final, subject-prominent versus topic-prominent (Greenberg 1963; Li & Thompson 1976; Dryer 2013) —and these defaults are acquired in childhood at no apparent cognitive cost. For an adult learner of an L2 that is typologically distant from her L1, however, the mapping must be performed deliberately, paying what we will call the Prefrontal Tax: the working-memory cost of restructuring an L1-shaped semantic plan into an L2-shaped surface form in real time (Levelt 1989; Hawkins 2004; Kormos 2006).

A structurally analogous problem has appeared in Large Language Model (LLM) prompting. Autoregressive Transformers (Vaswani et al. 2017) exhibit strong sensitivity to prompt linearization: identical semantic content, presented in different surface orders, yields measurably different output distributions (Lu et al. 2022; Sclar et al. 2024). When task-critical information is buried mid-prompt, models exhibit the Lost in the Middle effect (Liu et al. 2023); when prompt order conflicts with the model's positional priors, reasoning traces lengthen and final answers drift.

Both bottlenecks are, at heart, linearization-cost problems, and CFLT is the single remedy this paper develops for both: beyond placing the Core first, it constrains adjunct ordering by a convention motivated by listener-question priority, concreteness, and deictic recoverability. The protocol is intended to be language-neutral at the discourse layer while leaving each language's internal morphosyntax untouched.

The paper proceeds as follows. §2 defines the CFLT protocol and the Core/ground-frame partition. §3 motivates the protocol from human parsing and Transformer attention dynamics, with a deliberate primacy-versus-attention-sink disambiguation. §4 motivates the specific Reason → Space → Time adjunct ordering and labels it a convention with rationale, not a derivation. §5 illustrates operationalization across typologically distant languages. §6 reports a pilot two-part empirical evaluation that bears on the LLM-side prediction (P2) only; the human-side predictions (P1 psycholinguistic, P3 neural) and the typological / formal-semantic / human-AI-synchronization questions are reserved for the falsifiable research agenda in §7. §8 concludes. Readers from a cognitive-linguistic background should treat §6 as the LLM-side component of the broader research program introduced in §1.2; the human-side evidence is intentionally not in this paper, by the framing of §1.1.

The framework described here is maintained as an open commons (CC BY 4.0) at https://cflt.center; this paper is a derivative scholarly summary intended for academic peer engagement, not a substitute for the canonical specification.

1.1 A note on the cross-domain framing

The claim that SLA and LLM prompting share a "linearization cost" is a load-bearing cross-domain analogy and deserves an upfront caveat that we discharge here, before the protocol itself is defined in §2. Guest & Martin (2023) and the earlier critique tradition (McCloskey 1991) rightly warn that surface analogies between cognitive architectures and neural-network behavior routinely smuggle in unjustified mechanistic identifications. We accept the warning and adopt a deliberately weakened framing throughout the paper: CFLT does not claim that the two systems pay linearization cost through the same mechanism. The human "Prefrontal Tax" is mediated by the dorsolateral prefrontal cortex, left inferior frontal gyrus, and anterior cingulate cortex (DLPFC/LIFG/ACC; Abutalebi & Green 2007); the LLM "Lost in the Middle" curve is a behavioral positional-accuracy effect (Liu et al. 2023). What CFLT *does* claim is that a single normative intervention —Core-first linearization —happens to be useful in both systems for *partially distinct mechanistic reasons*, and that the human-side prediction (P1, §7.1) and the LLM-side prediction (P2, §7.3) are independently falsifiable. The pilot evidence in §6 bears on P2 only.

A second caveat applies on the human side alone. Slobin's (1996) "Thinking for Speaking" thesis is that linguistic encoding shapes which conceptual distinctions speakers habitually mark —i.e., the preverbal message (Levelt 1989's Conceptualizer stage) is not as fully language-neutral as Levelt's architecture suggests. If Slobin is right, CFLT's scaffold cannot fully *eliminate* L1-shaped Conceptualizer biases; it can only attenuate them by externalizing the linearization decision at production time. We accept this consequence and tighten the §7.1 (P1) falsification rider accordingly: the falsification condition for P1 is *attenuation*, not *elimination* —specifically, that CFLT-trained L2 learners should show measurably less L1-shaped time-fronting / topic-fronting than matched controls (a graded, not all-or-nothing, prediction), failing which the human-side claim is refuted. The multi-track structure of §7 (six sub-programs in §7.1–§7.6) is designed to allow the cross-domain claim to be evaluated component-wise rather than as a single up-or-down verdict; the Slobin-tightened P1 is one component of that evaluation.

1.2 CFLT as the natural-language layer of a broader program

This paper documents the natural-language layer of a broader research program built on a single organizing principle —*Core-then-Frame*: schema-enforced primary content preceding optional modifying content. CFLT applies this principle at the discourse layer, governing how human speakers and LLM receivers organize content for low-cost transmission. Whether the same Core-then-Frame ordering also helps carry discourse-level intent into the *structured tool-call interfaces* through which LLM agents act (e.g., Model Context Protocol (MCP) function schemas) is an open question we flag in §7.6 —not a contribution of this paper. The empirical contribution of this paper is the natural-language layer alone.

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2. The CFLT Protocol

CFLT is a discourse-level constraint on the relative order of two tiers of constituents: it places the speaker's salience anchor (the Core) at sequence-initial position and orders the circumstantial ground frame as `[Reason] → [Space] → [Time]`, while remaining deliberately silent about the internal grammar of each tier. It governs only the boundary between the two tiers and the order within the ground frame; everything inside a tier is delegated to the host language. Because it operates one level above clause syntax, a common misreading —that CFLT prescribes "verb-first" or "predicate-first" word order at the clause level —is mistaken: the protocol is neither, and §2.2 shows why the Core need not be a verb at all.

2.1 Two-tier structure

Every CFLT-conformant utterance is partitioned into two tiers (cf. Van Valin & LaPolla 1997):

- **Tier 1 —Event Nucleus (Slot 0).** The predicate together with its valence-bound participants (subject, object, instrument, beneficiary, recipient, accompaniment), nuclear- and core-level manner adverbials, and scope-internal operators (negation, modality, aspect, degree). Internal order within Tier 1 follows the target language's native syntax: SVO in English, SOV in Japanese, VSO in Modern Standard Arabic. The protocol does not touch Tier 1's internal grammar.
- **Tier 2 —Ground Frame (Slots 1–3).** Circumstantial adjuncts that situate the event in a world frame, ordered:

  `[Reason] > [Space] > [Time]`

The protocol governs only (i) the boundary between Tier 1 and Tier 2, and (ii) the internal order of Tier 2. The choice of morphosyntactic mechanism within each tier —case marking, prepositions, particles, coverbs, do-support, mood —is delegated entirely to the host language.

2.2 The "Core" as salience anchor

We define the Core as the salience anchor of the discourse —the constituent that the speaker is fundamentally "committing to" as the primary event, identity, state, or speech act. This characterization aligns the Core with Talmy's (2000) Figure in his Figure-Ground asymmetry (the salient, foregrounded entity whose path or location is at issue) and with Langacker's (1987, 2008) profile in cognitive grammar (the entity foregrounded for attention against a conceptual base). The CFLT Protocol places the Figure / profile at linear position 0 and the Ground / base in the subsequent slots.

Two routine misreadings must be ruled out explicitly:

- **Core ≠ "the most important constituent."** "Important" is a value judgment by the listener; the Core is a structural commitment by the speaker. In "*I went out, because the house was on fire,*" the most newsworthy information is *the house was on fire* —but the Core is *I went out*, because that is the speaker's primary commitment and the anchor against which the ground frame is interpreted. "Important" is not a defined technical term in formal linguistics, cognitive science, or NLP interpretability; adopting it as the CFLT term would conflate four independent dimensions (focus, salience, accessibility, surprisal) that information-structure theory (Krifka 2008; Gundel, Hedberg & Zacharski 1993) deliberately distinguishes.

- **CFLT's Core ≠ Role and Reference Grammar's Nucleus.** RRG (Van Valin & LaPolla 1997; Van Valin 2005) stratifies a clause as a nested hierarchy in which Nucleus (predicate alone) is contained within Core (predicate + arguments), which is contained within the full Clause (Core + Periphery). CFLT's Core aligns with RRG's *Core* layer plus a small portion of the Periphery (manner) and the scope-internal operator hierarchy of Cinque (1999), not with RRG's Nucleus. A reader who silently identifies CFLT's Core with RRG's Nucleus will incorrectly conclude that valence-bound arguments live in the ground frame.

2.3 Four types of Core, one protocol slot

Every well-formed CFLT utterance commits to one of four Core types in Slot 0:

Type	Predicate signature	Example (English CFLT form)	Theoretical anchor
Action	λx.λy. *event*(x,y, ...), eventive predicate, valence ≥ 1	*I didn't go out, because ...*	Vendler 1957; Dowty 1979; Levin & Rappaport Hovav 2005
Identity	λx. *be*(x, P), copular predication	*That girl is my sister, wearing ...*	Hengeveld 1992; Higgins 1979
State	λx. *state*(x), stative predicate	*I'm exhausted, because ...*	Carlson 1977; Maienborn 2005
Request	DIR(s, h, P), directive speech act	*Could you pass the salt, please ...*	Searle 1969, 1975; Sadock & Zwicky 1985

The selection of Core type is a semantic decision by the speaker ("what am I really trying to say?"). The placement of Core in Slot 0 is the protocol CFLT enforces. The four types share a single protocol slot precisely because the protocol is constituent-type-agnostic at the discourse layer; what fills Slot 0 internally is type-specific and language-specific.

2.4 Layer-by-layer universality

Whether CFLT is "universal"depends on which layer one looks at. The table below states the canonical position:

Layer	Content	Universal?	Role of any specific language
Layer 1: Protocol	Core in Slot 0; ground frame ordered R > S > T	**Yes —within the surveyed typological range**	No language is privileged.
Layer 2: Slot semantics	Which functional question each slot answers (Why / Where / When)	**Yes —within the surveyed typological range**	These are functional categories, not surface syntax.
Layer 3: Event-nucleus internal assembly	How predicate + valence + manner are arranged inside the Core	**No —fully language-specific**	Each language uses its own native syntax.
Layer 4: Boundary edge cases	Whether "*with X*" / "*in X*" / etc. attach inside Core or to a ground-frame slot	**Mostly universal, with language-specific edge cases**	English may serve as a verification anchor but not as judge.

CFLT's universality claim is restricted to Layers 1 and 2. The other two layers explicitly delegate to language-specific machinery, and that delegation is what makes the universality claim defensible. The position is deliberately weaker than the strong-UG framing of earlier drafts and is compatible with usage-based / construction-grammar accounts (Tomasello 2003; Goldberg 1995, 2006), with the anti-UG critiques of Christiansen & Chater (2008) and Evans & Levinson (2009), and with the typological-functional tradition of Croft (2001) —the central claim is that Levelt's (1989) preverbal message formation is broadly shared across speakers, not that an innate language-specific Universal Grammar is required. The Layer-2 slot semantics (Why / Where / When) corresponds closely to the Natural Semantic Metalanguage primes BECAUSE / WHERE / WHEN (Wierzbicka 1996), which Wierzbicka argues are lexicalized in every documented language —providing an independent typological motivation for treating these three slot-functions as functionally universal even when their surface morphosyntactic realization varies.

2.4.1 Typological challenges to the protocol-layer universality

Because that universality claim is strong, we stress-test it here against the typological patterns most likely to break it —a robustness check a reader who accepts the scope restriction (§2.4) can skip. A typologist will point to several documented patterns that appear to contradict the claim; we engage three of the strongest:

- **Yup'ik discourse-driven fronting (Mithun 1992).** Yup'ik regularly fronts pragmatically salient constituents — newness, topic shift, contrast —rather than event-anchored Cores. The clause-initial position is determined by discourse pragmatics, not by event identification.
- **Evidential-first languages (Aikhenvald 2004).** In Tariana, Tuyuca, Quechua, and Aymara, the obligatory leftmost element is an evidential marker encoding the speaker's information source (visual / non-visual / inferential / reportative / hearsay). Arguably, the evidential is the salience anchor in these languages.
- **Tagalog topic-prominence (Foley & Van Valin 1984).** The *ang*-marked topic is grammatically privileged and need not coincide with the event anchor.

Four points of response, in order of strength:

1. **Scope restriction (already canonical).** CFLT's typological scope is the surveyed range (§2.4 Layers 1–2 apply to Indo-European, Sino-Tibetan, Japonic, Koreanic, Afro-Asiatic). Yup'ik (Eskimo-Aleut), Tariana (Arawakan), Tagalog (Austronesian) are outside it. The universality claim does not extend to them; the rest of this section explains why this is not just retreat.
2. **Salience anchor ≠ event anchor —with an independent test, not a tautology.** Broadening "Core"beyond the action-verb type (§2.2's four types) risks circularity: if the anchor were just "whatever sits at position 0,"the protocol would be unfalsifiable. To prevent this we pre-commit two *position-independent* operational tests that identify the salience anchor without reference to where it currently sits, making the protocol-layer claim falsifiable per language —a systematic mismatch between the tests and a language's default linearization is grounds for restricting the claim, not for redefining the anchor.[2]
3. **Normative protocol vs. descriptive typology.** Even if Yup'ik's *current* default linearization is pragmatically driven, this is a fact about Yup'ik's descriptive grammar, not a verdict on whether a Yup'ik L2 learner of English would benefit from a CFLT scaffold. The Layer-1/2 universality claim concerns the L2-production scaffold layer, not the L1's descriptive surface grammar.
4. **Terminological mismatch.** The word "Core"has eventive resonance and may mislead readers from non-event-prominent traditions. A possible future expansion of the taxonomy is a fifth Core type —Evidential/Stance Core —to cover the evidential-first languages explicitly. Until that work is done, we restrict the Layer-1/2 universality claim to the surveyed typological range and treat evidential-first and pragmatically-ordered languages as principled extensions pending empirical investigation, not refuting counterexamples.

This response does not defend CFLT *outside* the surveyed range; it defends the *restricted* version against the *general* refutation. The deeper engagement with the anti-UG opposition (Tomasello 2003; Christiansen & Chater 2008; Evans & Levinson 2009; Newmeyer 2005) and with the Mithun/Aikhenvald typological literature is maintained in the canonical foundations document (cflt.center/foundations/linguistics §6.2).

[2]The two tests: **(T1) speaker-commitment substitution** —the anchor is the constituent whose lexical substitution most changes *what proposition the speaker is asserting*, holding the rest fixed (substituting a place adverbial yields the same event under a different scene-frame; substituting the anchor yields a different commitment); **(T2) listener-question** —the anchor answers the listener's first natural question after the speaker stops mid-utterance ("what is the speaker doing / asserting / committing to?"), independent of which constituent sits at position 0. Falsification rider: if (T1) and (T2), applied to unmarked declaratives in a non-event-prominent language, identify an anchor other than the one at position 0 in that language's default order, then either CFLT's prescription is mis-aligned and the universality claim is refuted for that language, or the language is a candidate for the genre-conditional / fifth-Core-type extension (point 4) —admissible only if the mismatch is systematic, not item-by-item rescue.

3. Cognitive and Computational Mechanics

§2 defined *what* the protocol is; this section asks *why* it should lower linearization cost —first in human sentence production (§3.1), then in LLM attention dynamics (§3.2). The two mechanisms are distinct, consistent with the joint-not-unitary framing of §1.1.

3.1 Human parsing: from preverbal message to incremental processing

Levelt's (1989) production architecture distinguishes three stages: Conceptualizer (preverbal message formation, language-neutral), Formulator (grammatical and phonological encoding, language-specific), and Articulator (motor execution). The essential point here is that the Conceptualizer's output is language-neutral —the semantic core of an intended event exists before any L1- or L2-specific surface form.

CFLT's pedagogical claim is grounded here: if the preverbal message is language-neutral, then training the learner to linearize the preverbal message in a fixed Core-First order *before* entering the Formulator stage decouples conceptual structuring from L1 surface grammar. Both L1 and L2 formulation then become token-substitution exercises over the same linearized scaffold, rather than full structural restructuring.

This account is further motivated —by analogy, not derivation —by Hawkins's (1994, 2004) Early Immediate Constituents (EIC) principle, formalized as Minimize Domains, under which the human processor prefers structures that allow the rapid identification of phrasal heads, shortening the Constituent Recognition Domain (CRD). CFLT applies this sentence-level principle at the discourse layer: placing the Core at Slot 0 is predicted to minimize the recognition domain for the anchoring constituent of the discourse, reducing the look-ahead buffer load on working memory.[3] For learners coming from a head-final or modifier-heavy L1 (e.g., Mandarin pre-nominal relative clauses; Japanese clause-final verbs), this inverts the cognitively expensive Modifier Trap at the discourse layer: the learner discharges working-memory contents incrementally, treating subsequent adjuncts as low-priority refinements that can be appended without revising prior commitments.

3.2 LLM Transformer attention: primacy, ***not*** attention sinks

In plain terms: a model that meets the task in its first tokens spends less effort hunting for it later, so CFLT puts the task-bearing Core where the model already attends most —the prompt prefix. Making that precise requires a careful disambiguation, because two distinct mechanisms cause early-position over-attention in Transformer LLMs and are routinely conflated.

- **Attention sinks (Xiao et al. 2024).** Because the softmax denominator must sum to 1, attention "leaks"into the very first tokens of a sequence regardless of their semantic content. Xiao et al. explicitly note these tokens are *"not being semantically important"*—the sink is a softmax-stability artifact, not a signal of semantic priority. Modern systems typically reserve the beginning-of-sequence token `<bos>` as a dedicated sink slot.
- **Primacy / positional bias.** Independently, early tokens are attended more heavily by every subsequent token's queries because causal masking compounds early influence over depth. This effect *does* favor semantically rich content placed early. The Lost in the Middle finding of Liu et al. (2023) —a U-shaped accuracy curve in long-context tasks favoring beginning and end over middle —has, in its *beginning*-favoring half, a document-scale manifestation of this primacy effect (the *end*-favoring half reflects a distinct recency effect, not primacy).[4]

CFLT exploits primacy, not the sink. When the Core is buried in a prompt, the model's early conditional distribution is high-entropy and downstream tokens inherit that uncertainty —reasoning-capable models lengthen their internal traces because they are, in effect, still searching for the task. Placing the highest-information constituent at the prompt prefix

[3] Hawkins's EIC is defined and validated at the scale of constituents *within* a clause; its application to the order of discourse-level units (Core vs. ground frame) is an extrapolation CFLT adopts as motivation —in the same way the Liu et al. (2023) document-scale result motivates but does not prove the sentence-scale prediction of §3.2. The discourse-level load reduction is a predicted effect (P1, §7.1), not a measured one.

[4] Liu et al.'s result is at document scale (information distributed across 10–30 retrieved documents); applying it to sentence scale —clause order within a single utterance —is an analogy, not a measured result. The sentence-scale prediction is treated as empirical content requiring its own validation (§6, §7.2), motivated by primacy + sink dynamics rather than proved by Liu et al.

collapses the early branching factor and stabilizes the autoregressive trajectory. The mechanism interacts naturally with Chain-of-Thought (CoT) prompting (Wei et al. 2022): CFLT contributes stable Slot-0 anchoring for the *task operator*, while CoT contributes the intermediate reasoning structure CFLT does not specify; the two are complementary rather than competing. (This mechanistic claim is predicted, not measured here —the attention-rollout and causal-attribution tests required to verify it are in §7.3.) Putting CFLT's Core at position 0 does *not* "consume"the sink; it occupies the high-attention prefix region immediately after `<bos>`.

The primacy advantage is nonetheless task-dependent, not a universal first-position favoritism. Pezeshkpour & Hruschka (2024) find that multiple-choice accuracy shifts by tens of percentage points with option order, the *direction* of the bias differing across models; Berglund et al. (2024, "The Reversal Curse") show that order-sensitivity is entangled with the learned representations themselves, not just with prompt-time formatting. Two consequences for CFLT follow: (a) the buried-decision condition (Level 4) can interact with model-specific properties rather than uniformly benefiting from Slot-0 placement —one model regresses while the others saturate, reported and analyzed in §6.5; and (b) the stronger *CFLT-as-alignment-constraint* upgrade (§7.3) is harder than CFLT-as-prompt-protocol, because the Reversal Curse implies training-time exposure may not yield inference-time robustness.[5]

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4. The Adjunct Ordering: Convention with Rationale, Not Derivation

The claim that the Core occupies Slot 0 is *motivated* by several convergent rationales drawn from adjacent traditions (Talmy's Figure-Ground asymmetry; Hawkins's EIC; Levelt's preverbal message; primacy effects in causal-attention Transformers; Gricean Relevance) —motivations and analogies that make Core-first a well-grounded design choice, not independent confirmations that it lowers production or processing cost, which is the empirical question of §6–§7. The claim that the internal order of the ground frame is Reason > Space > Time is a different kind of claim and deserves a separate, more careful framing: it is a convention selected from the 3! = 6 permutations of the three slots, motivated by three rationales —none of which alone constitutes a derivation of R-S-T as the unique optimum.

(i) Listener-question priority (Gricean Relevance). After receiving the Core ("what happened"), the listener's most discourse-coherent next question is "why?"—the cause typically renders the event interpretable. Place and time are scene-locators that the listener can usually defer or infer from context. Reason therefore sits closest to Core.

(ii) Concreteness ladder. Spatial information is more concrete and perceptible than temporal information, which is more deictic and abstract. Working memory benefits from a concrete-to-abstract progression: hearing *where* helps the listener mentally place the event before the more abstract *when* binding closes the scene. We treat this rationale as provisionally defensible within the surveyed typological range, and flag it as a separable open question whether listener-question priority (rationale (i)) alone suffices to motivate R-S-T without it.[6]

(iii) Deictic recoverability. In conversational contexts, time often has a recoverable default ("now"or "the time being discussed"); placing it last allows omission without information loss in many contexts.[7]

These are engineering arguments. The strongest motivating consideration on the other side comes from the discourse-analysis literature: Reinhart (1984) shows that temporal structure is central to the coherence and organization of narrative texts. This motivates —though Reinhart does not herself propose a slot order or directly compare *when* against *why* —a CFLT-internal hypothesis that for narrative-genre production a Core → Time → Reason → Space (R-T-S) order

[5] A narrower scope note: prompt-order sensitivity (Lu et al. 2022; Sclar et al. 2024) places CFLT's contribution on the format-and-order axis, not on content correctness —fixing both the schema and the linear order of the speaker's commitment is the joint discipline. Min et al.'s (2022) partial counter-result is a related caution from an adjacent setting: in few-shot in-context learning, the *correctness of the input–label mappings* in demonstrations matters surprisingly little, while format, label space, and input distribution carry much of the effect. (Min et al. study demonstration *properties*, not constituent order; we do not cite them for an order effect.) The lesson CFLT takes from it is that a measured CFLT gain could come from communicating task *format* rather than from semantic linearization —a confound the evaluation must isolate (§6.5, §7.3).

[6] The concreteness rationale is not culture-neutral. The conceptual-metaphor literature shows that humans recruit spatial structure to talk about time (Lakoff & Johnson 1980), but the *direction* of the mapping is language-specific (Boroditsky 2000 on Mandarin vertical time-axis; Casasanto & Boroditsky 2008 on asymmetric space-to-time priming; Núñez & Sweetser 2006 on Aymara's reversed time deixis). It therefore generalizes only to languages where the dominant time-talk strategy is spatially-mediated; where time is encoded primarily by grammatical tense/aspect/evidentiality, the rationale weakens.

[7] Levinson (1983) treats speaker / time / place as three primary deictic axes without arguing one is more recoverable than another; the recoverability claim here is a CFLT-internal observation about non-co-present L2 discourse, not a theorem of deixis theory.

might be preferable. CFLT selects R-S-T for its two targeted use cases (L2 conversational/expository pedagogy and LLM prompt stability), where the listener-question and concreteness rationales jointly outweigh narrative-temporal anchoring; narrative L2 production is a recognized boundary case where the R-T-S alternative —a CFLT-generated competing condition, tested in §7.2 —may perform better and remains an open empirical question.

Operational consequence. *Core in Slot 0* is the protocol's fixed default —the one design commitment CFLT does not treat as negotiable, because it is the commitment that makes the protocol a protocol —and it is motivated by the convergent rationales enumerated at the head of this section. Its *design* status (fixed) is distinct from its *empirical* status (open): whether Core-first actually lowers human or machine linearization cost is the hypothesis under test in §6–§7, not a settled result. CFLT's weaker claim —*R then S then T* —is, by contrast, negotiable even as a design choice: it is a documented convention with stated reasons and may be revised if empirical evaluation shows another permutation outperforms it for a given genre or language pair.

4.1 Coverage: how CFLT compresses Halliday's nine circumstance roles

Systemic Functional Linguistics (Halliday & Matthiessen 2014) decomposes circumstantial adjuncts into nine semantic roles. CFLT's three ground-frame slots are a structured compression of this taxonomy:

Halliday role	CFLT location
Extent (duration, frequency)	Slot 3 [Time]
Location: place	Slot 2 [Space]
Location: time	Slot 3 [Time]
Manner: quality (e.g., *slowly*)	Inside Core (event nucleus)
Manner: means (e.g., *by phone*)	Inside Core (instrument)
Manner: comparison (e.g., *like X*)	Inside Core (manner sub-type)
Cause: reason / purpose / condition / concession	Slot 1 [Reason]
Cause: behalf (beneficiary, *for X*)	Inside Core (valence-bound)
Accompaniment (*with John*)	Inside Core (valence extension)
Matter (*about X*) / Angle (*according to X*)	Slot 2 [Space] (abstract domain)

The compression follows the two-tier model exactly: roles internal to the event (how, with-what, with-whom, for-whom) collapse into the event nucleus, while roles framing the event (why, where, when, in-what-respect) populate the ground frame. The compression we adopt is functional-level: Location:place, Matter, and Angle all answer the listener's "in what domain (physical, semantic, perspectival) does this hold?"question, so the compression is *functional-level lossless* and *SFL-experiential-level lossy* —a deliberate trade-off, because CFLT is a discourse-level production scaffold, not an SFL replacement.[8]

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5. Operationalization

With the protocol defined (§2), motivated from human and machine parsing (§3), and its adjunct ordering and coverage set out (§4), we now show it in operation across typologically distant languages and in human-to-LLM prompting. CFLT is a normative scaffold, not a descriptive claim about natural-language defaults. Its operational value is that it converts structural transformations into lexical substitutions: the learner or prompt author no longer chooses *where to place* a constituent, only *what to place* in each protocol slot.

Example A —Mandarin (topic-time-first L1) to English (head-initial L2): - L1 default: 昨天 (Time) 下雨 (Reason)，我没出去 (Core)。- CFLT pivot: [Core: 我没出去] → [Reason: 因为下雨] → [Space: 在家] → [Time: 昨天] - L2 output: "*I didn't go out, because it was raining, at home, yesterday.*"

[8] A reader familiar with SFL will rightly object that Halliday treats Matter (*about X*) and Angle (*according to X*) as distinct from Location:place precisely because the abstract/concrete cut is theoretically loaded; the lossy compression is adopted with that objection in view. A residual edge case (Halliday's *Role*, e.g., *acting as a teacher*) is handled case-by-case in the project's slot-disambiguation reference. The full SFL treatment, including the Matter/Angle classification basis, is maintained at cflt.center/foundations/linguistics §4.4.1.

Tier 1 (*I didn't go out*) is assembled by English's native morphosyntax —do-support, SVO order, post-verbal negation; the speaker no longer has to *decide* whether to front time or topicalize reason. The protocol fixes that choice.

Example B —Same pivot to Japanese (head-final L2): - CFLT pivot: identical - L2 output (spoken / right-dislocated register): 出かけなかった、雨が降ったから、家で、昨日。

Japanese's Tier 1 internal order (SOV) is preserved by its native syntax; only the boundary between Tier 1 and Tier 2 and the order within Tier 2 are governed by the protocol. The protocol survives the typological change even though the surface order of Tier 1's internal arguments does not.

Example C —Human-to-LLM prompt (English): - Natural prose: *"Yesterday, at the cafe across from the station, because the wifi was unstable, please summarize the attached document into three bullet points."* - CFLT-reordered: *"Please summarize the attached document into three bullet points, because the wifi was unstable, at the cafe across from the station, yesterday."*

The semantic content is identical; the second form positions the Core directive (*summarize ...three bullet points*) at the prompt prefix, where it benefits from primacy attention.

A worked five-language demonstration (English, Mandarin, Japanese, Korean, Modern Standard Arabic) spanning Indo-European, Sino-Tibetan, Japonic, Koreanic, and Afro-Asiatic typologies, with the native morphosyntactic mechanism (negation, aspect, argument marking) explicitly identified for each, is given in the project's canonical core-concept reference. We present this as an *illustrated feasibility check*, not as evidence of cross-linguistic invariance: it shows that a single protocol-layer order *can be constructed* in five typologically distant languages, each assembling its event nucleus with a completely different morphosyntactic toolkit. Whether that order is also unmarked, natural, or pedagogically beneficial in each language is a separate empirical question requiring native-speaker acceptability judgments and corpus frequencies not yet collected (§7.2); authored examples establish constructability, not naturalness.

The protocol has been implemented operationally as a human-bilingual-education application (see Data and Code Availability) providing four surfaces: a Logic Transformer that restructures natural-language input into CFLT JSON, scenario-based course generation for English↔Mandarin adult learners, multi-turn roleplay with per-turn CFLT-compliance feedback, and progress analytics over local learning logs. The Logic Transformer surface corresponds operationally to the Part I benchmark in §6.4 —the same task, on the same dataset schema, packaged for end-user interaction.

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6. Pilot Empirical Evaluation

To assess CFLT's computational claim —that Core-initial linearization stabilizes LLM behavior —we conducted a controlled two-part study (May 2026). All prompts, raw API responses, scoring scripts, and aggregate reports are released alongside this paper (see Data Availability) and are reproducible from a single command. This is a pilot study; its limitations are made explicit in §6.5, and we frame the results as suggestive rather than confirmatory.

6.1 Design

The study uses a single shared dataset of 24 cases distributed as 4 difficulty levels × 2 languages × 3 scenarios (12 English + 12 Mandarin items). Each case provides:

- `utterance_control` —natural-prose phrasing, typically reason-first or topic-first;
- `utterance_cflt` —the **same lexical content** reordered as `Core → Reason → Space → Time`; no markup tokens, no compression;
- `ground_truth` —canonical extraction in a fixed schema (`action / target / location / time`).

Levels were designed to probe distinct properties:

Level	Probe
Level 1	Single action, 2 fields —ceiling probe; both arms expected to saturate.
Level 2	Action + location + time, 4 fields —moderate load.
Level 3	Level 2 plus *distractor* clauses (red herrings, ongoing context) —primary test of "Lost in the Middle" mitigation.
Level 4	Multiple candidate actions, the *decided* action buried at end-of-utterance in control —primary buried-decision primacy test.

Part I —Logic Transformer Benchmark asks whether the CFLT transformation can be performed *by an LLM*. Each case's `utterance_control` is fed to a transformer LLM under a fixed system prompt (no ground-truth values, only the output schema). Three structural-compliance metrics are scored automatically —SC (slot order Core→Reason→Space→Time), SR (Core slot retains the subject), IV (inferred slots carry justifying suggestions) — and a DX metric measures downstream extraction accuracy when the transformer's output is fed back to a separate extractor LLM.

Part II —Control vs. CFLT Extraction is the primary ablation. Each case's `utterance_control` and `utterance_cflt` are fed independently to the same extractor LLM with the same schema-only system prompt. The two arms differ only in clause order. Each arm is repeated N=3 times at `temperature=0` (provider defaults where the API rejects the parameter), giving 24 cases × 2 arms × 3 runs = 144 API calls per model.

Methodological guardrails enforced by the evaluation harness:

- No ground-truth leakage: the system prompt declares only the schema (`action / target / location / time`), never the canonical values.
- The two arms share identical lexical content; the manipulation is **order only**.
- Per-case variance is reported (mean ± std across the 3 runs).
- `prompt_tokens` and `completion_tokens` are reported separately, never merged into a single "gain"number, because they respond to different mechanisms.
- Surface-form synonyms (e.g., `"6 PM today"` ≡ `"today 6 PM"`; *5* 楼 ≡ *5* 楼) are dataset-driven and auditable, not hidden in scoring code.

Five current-generation models were evaluated in Part II, spanning four distinct output regimes: `openai/gpt-5` (concealed reasoning), `google/gemini-3-flash-preview` (short output), `qwen/qwen3.5-plus` (visible chain-of-thought), `deepseek/deepseek-v4-pro` (visible chain-of-thought), and `anthropic/claude-sonnet-4-6` (short output). The five models come from five different commercial providers; the cross-provider coverage strengthens the generalizability of the resulting cross-model pattern.

6.2 Part II Results —Accuracy

We report non-ceiling Δ-accuracy: the accuracy delta restricted to levels where the control arm has not already saturated (≥ 90%), because on saturated levels CFLT has no headroom and including them dilutes the verdict.

Model	Informative levels	Non-ceiling Control	Non-ceiling CFLT	Δ
GPT-5	Level 3	61%	100%	**+38.9 pp**
Gemini 3 Flash	Level 3	78%	100%	**+22.2 pp**
Qwen3.5-Plus	Level 2, Level 3	75%	97%	**+22.2 pp**
DeepSeek V4 Pro	Level 3, Level 4	69%	86%	**+16.7 pp** (Level 3 alone: +44pp; Level 4 −11pp regression drags aggregate, see §6.5)

Model	Informative levels	Non-ceiling Control	Non-ceiling CFLT	Δ
Claude Sonnet 4.6	Level 2, Level 3	78%	89%	**+11.1 pp** (Level 3 alone: +28pp; Level 2 −6pp noise drags aggregate)

The Level-3 (distractor-heavy) effect is consistent and large across all five models: control accuracy ranges from 56% to 78% (mean 65.6%), CFLT accuracy saturates at exactly 100% for every model. The five-model Level 3 universality is the primary finding of the study: the primacy advantage is observed across four distinct output regimes (concealed reasoning, visible chain-of-thought, two short-output non-reasoning models) and five different commercial providers.

The Level-4 (buried-decision) effect splits the models: four of the five sit at or near ceiling under both arms (GPT-5 100%/100%, Gemini 94%/94%, Qwen 94%/92%, Claude 100%/100%), and only DeepSeek V4 Pro shows a −11 pp regression (83% → 72%). The Level 4 cases place multiple candidate actions before a final decision; the control's natural narrative sequence (alternatives → final choice) provides a potential reasoning scaffold that CFLT front-loading removes. We retain this single-model anomaly because reporting it is intrinsic to the falsifiability discipline (§7.3); the competing interpretations are weighed in §6.5.

6.3 Part II Results —Token Cost

Token effects split cleanly along the reasoning-versus-non-reasoning axis:

Model	Prompt tokens Δ	Completion tokens Δ	Reasoning trace?
GPT-5	−1.5%	+0.7%	concealed (no exposed trace)
Gemini 3 Flash	−1.4%	+1.4%	no
Claude Sonnet 4.6	−1.2%	+0.9%	no
Qwen3.5-Plus	−1.1%	**−38.4%**	yes (visible chain-of-thought)
DeepSeek V4 Pro	−1.1%	**−12.5%**	yes (visible chain-of-thought)

Prompt tokens move trivially in either direction (≤ 1.5%), as expected when both arms carry identical lexical content. Across the five models, the cluster structure is unambiguous: completion tokens drop substantially only on the two models with visible reasoning traces (Qwen, DeepSeek), while the three short-output or concealed-reasoning models (GPT-5, Gemini Flash, Claude Sonnet 4.6) all show completion-token Δ within ±1.5% (pure noise). This clean two-cluster split is consistent with the §3.2 prediction that an early Core constraint collapses the branching factor of the model's internal search —*which can only show a token effect when that search is externalized in the visible completion stream*. An illustrative case: Qwen's `ZH_L3_03` arm shows the control trace at 17,092 completion tokens (long confused reasoning, wrong final answer) versus the CFLT trace at 2,821 tokens (focused reasoning, correct) —a 6× reduction on a single case. The three short-output / concealed-reasoning models show no meaningful completion-token effect, which is neither evidence for nor against the protocol's claim, because no externalized internal search is being shortened.

A clarifying note: the original CFLT documentation projected a 30–50% prompt-token savings under a "compressed CFLT"formulation with explicit slot labels and `NULL` fillers. The current dataset uses lexically identical content in both arms (the manipulation is purely order), so prompt compression is not testable by design. To test the prompt-economy claim directly, a third "compressed CFLT"arm is required; this is reserved for the registered confirmatory study (§7).

6.4 Part I Results —Logic Transformer Feasibility

Part I (using `deepseek/deepseek-v4-pro` as the transformer and `openai/gpt-5` as the downstream extractor, N=1):

- Structural compliance: SC = SR = IV = **100%** across all 24 cases. The transformer reliably produces well-formed CFLT JSON with correct slot order, subject preservation, and inference suggestions.
- Downstream extraction accuracy (DX): Level 1 = 100%, Level 2 = 83%, Level 3 = 67%, Level 4 = 100%; overall = 88%.

- The Level 3 DX (67%) is comparable to the Part II Level 3 control baseline (61–78%) but well below the human-authored CFLT ceiling (100% in Part II). On inspection, the three Level 3 DX failures are stochastic (N=1) rather than systematic; a v3 snapshot of the same prompt achieved 23/24 = 96%.

Implication. CFLT is currently a usable preprocessing protocol when authored by a human, and a *structurally* implementable protocol when authored by an LLM —but LLM-authored CFLT does not yet preserve enough content fidelity at Level 3 to recover the full Part II accuracy gain. Closing this gap (e.g., by fine-tuning a Logic Transformer on CFLT-conformant data) is a concrete research-agenda item (§7.3).

6.5 Limitations

We enumerate limitations explicitly because the headline number —100% accuracy on Level 3 across all five models —is larger than is plausible for a mature evaluation and reflects properties of the pilot set as much as a real ceiling.

- **Sample size.** 24 cases × 3 runs × 2 conditions × 5 models = 720 Part II trials is adequate for descriptive cross-model comparison and is the empirical base for the five-model Level 3 universality claim, but it is still below the threshold for confirmatory mixed-effects inference at the per-case level. A pre-registered larger-scale evaluation ($N \geq 10$ runs per case; per-case-author replication) is the next step in §7.
- **Ceiling saturation.** Level 1 control accuracy is at ceiling for all five models; Level 2 saturates for three (GPT-5, Gemini, DeepSeek), shows mild positive noise on Qwen (+6 pp), and is informative-with-negative-noise on Claude (−6 pp, see next bullet); Level 4 saturates for four of five models, with only DeepSeek showing the −11 pp regression discussed below. Level 3 is the only level that is reliably informative across all five models. Future case design must target the regions where current frontier models actually fail.
- **Claude Level 2 noise observation.** Claude Sonnet 4.6 is the only model where Level 2 showed a non-zero aggregate (83% → 78%, −6 pp), but per-case inspection shows this is sampling noise, not a stable regression: strong single-case positives and negatives mostly cancel, and at N=3 runs per arm the −6 pp sits within sampling noise.[9] A confirmatory study with $N \geq 10$ per case is required to resolve the direction.
- **Level 4 null/negative result.** Of the five models, only DeepSeek V4 Pro shows an informative Level 4 regression (−11 pp); the other four are at or near ceiling on Level 4 under both arms (GPT-5 100%/100%, Gemini 94%/94%, Qwen 94%/92%, Claude 100%/100%). The reading most consistent with the data is a DeepSeek-specific model × item-type interaction, not a general primacy failure (four of five models show no Level 4 issue). To keep that from sliding into ad-hoc rescue, we pre-commit anti-rescue criteria for the confirmatory study: the redesigned Level 4 set must hold embedding-distance between the buried decision and the distractors above a pre-registered threshold, syntactically anchor the decision (e.g., “我决定”, “I decide”) rather than merely position it last, and be re-authored by ≥ 2 independent annotators —if the regression survives all three it is locked in as a model × item interaction, and if it disappears under any it is refuted.[10] The primacy mechanism (§3.2) stays consistent with the five-model Level 3 result and the four-of-five Level 4 saturation; operationally, deployments on DeepSeek for multi-option decisions should A/B-test CFLT before committing.
- **Model coverage.** Five models across five commercial providers is the cross-provider base for the Level 3 universality claim; the four-of-five Level 4 saturation result is also drawn from this set. Stronger generalization claims (and tighter resolution of the DeepSeek Level 4 anomaly) require ≥ 8 frontier models including at least one from each major architecture family, with a planned re-run when checkpoints update.
- **Construct validity of “distractor density.”** The four difficulty levels were operationalized at case-construction time by a single author. A principled, embedding-distance-based difficulty metric should replace this in the confirmatory study.
- **Statistical inference.** With N = 24 cases per condition we report descriptive statistics and per-case mean ± std only. Bootstrap CIs, mixed-effects modeling with case and language as random effects, and pre-registered hypothesis tests are deferred to the confirmatory study.

[9] Per case: ZH_L2_01 went 0% → 100% (+100 pp), ZH_L2_02 went 100% → 0% (−100 pp), EN_L2_03 showed −33 pp; the remaining three Level 2 cases were saturated on both arms. The −6 pp aggregate is the algebraic residual after these cancel.

[10] Three readings were weighed against the cross-model evidence. (a) *Primacy over-strong as a unified explanation —weakly supported*: it predicts the Level 4 regression should recur wherever Level 4 is informative, yet four of five models show none; a weaker surviving form (primacy locally overridden when the natural narrative order is itself a strong reasoning scaffold) rests on N=1 model. (b) *DeepSeek-specific model × item interaction —most consistent with the data* (guarded by the anti-rescue criteria above). (c) *Reasoning models compensate via internal search —not supported*: it predicts the regression should track reasoning-trace length, but Qwen (longest traces) shows Level 4 noise (−3 pp), DeepSeek (shorter traces) shows −11 pp, and Claude (no visible trace) shows zero; not refuted given N=5, but deprioritized.

- **Language coverage.** The dataset is bilingual (English + Mandarin), which is a strict improvement on the English-only baseline implicit in much prompt-engineering work; but it is still a thin sample of the world's typological space. Extending to head-final languages (Japanese, Korean) and to a VSO language (Arabic) is a near-term priority.
- **Model-version drift.** Frontier LLMs update on month-scale timescales; the May 2026 checkpoints may not replicate on subsequent releases. The released code re-runs the entire study in a single command for verification.
- **Scoring is automated, not human-rated.** Binary extraction was scored programmatically against a dataset-driven synonym table; we did not manually verify a held-out sample for inter-rater agreement against the automated judge. This is a future-work item.

We therefore characterize these results as suggestive —with one informative null result (Level 4) that should constrain the next iteration of the theory —rather than as confirmatory evidence for CFLT-in-general.

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7. Open Questions and Research Agenda

CFLT's value as a theoretical claim rests on whether it makes falsifiable predictions across cognitive, typological, and computational dimensions. We enumerate the open questions that, in our view, define a multi-phase research program. (The canonical formulation is in the project's empirical-agenda reference; this section summarizes and re-orders for this paper's scope.)

7.1 Psycholinguistic —Production-speed advantage under L2 conditions (P1)

Claim. Adult L2 learners taught the CFLT unmarked default will produce sentences with higher fluency (words per minute) and lower hesitation rate than learners taught a free-order baseline, controlled for vocabulary and grammar coverage.

Methodology. Between-subjects intervention; two groups of intermediate L2 learners receive identical lexical/grammatical content for N weeks; one group additionally drills the CFLT four-slot template. Measure words per minute in elicited monologue, hesitation pauses ≥ 250 ms (Pawley & Syder 1983), and error rate.

Falsification condition. Two graded criteria, both must hold for the claim to survive: (i) the CFLT-trained group shows a statistically significant fluency advantage at $p < .05$ across at least two replication sites; and (ii) the CFLT-trained group shows measurably reduced L1-shaped fronting (e.g., for Mandarin-L1 / English-L2, reduced rate of unmarked time-initial declaratives relative to matched controls) —this is the Slobin-tightened criterion (§1.1), and reflects that CFLT is predicted to *attenuate*, not eliminate, L1-shaped Conceptualizer biases. If (i) holds but (ii) does not, the fluency benefit is real but not attributable to L1-bias attenuation; if (ii) holds but (i) does not, the bias is reduced without translating to production-time benefit; either single failure refutes the joint P1 claim as stated.

Adjacent evidence. VanPatten's (1996, 2004) Processing Instruction work supports the existence of a "default schema" benefit methodologically, though not directly for CFLT. Kormos (2006) places the L2 production bottleneck at the Formulator stage, predicting a CFLT effect mechanistically. DeKeyser (2007) provides the declarative-to-procedural skill-acquisition curves into which CFLT should fit. Slobin (1996) defines the upper bound on the predicted effect — CFLT attenuates rather than eliminates the L1 bias.

7.2 Typological —Is R > S > T the unique optimum?

The R-S-T inner order is a convention (§4), not a derivation. Two extensions are needed:

(a) Quantitative corpus work across at least one language per major family (Mandarin, Japanese, Arabic, Swahili, Quechua). *Falsification condition*: in unmarked declaratives with ≥ 2 ground-frame adjuncts co-occurring, the relative-order frequencies (R-S, R-T, S-T pairs) should show R preceding S, R preceding T, and S preceding T in ≥ 60% of attested co-occurrences in ≥ 4 of the 5 families. If the frequency floor is missed in ≥ 2 families, the unmarked-default claim of R>S>T is refuted as a cross-linguistic generalization within the surveyed typology, and CFLT should retreat to a per-language convention.

(b) A registered narrative-genre test of the Time-first hypothesis —a CFLT-generated competing condition (R-T-S) motivated by Reinhart's (1984) account of temporal structure in narrative organization, not a production order Reinhart herself prescribes —with the falsification condition that if Time-first reliably outperforms R-S-T for narrative L2 production across ≥ 2 languages (measured by fluency, hesitation rate, and listener comprehension at $p < .05$), CFLT should adopt a genre-conditional protocol —R-S-T for expository/conversational registers, R-T-S for narrative.

7.3 Computational —LLM attention concentration and protocol-vs-alignment (P2)

Pilot status. The pilot study (§6) treats CFLT as a prompt-time intervention; the Level 3 finding (100% accuracy across all five models) is suggestive but not yet confirmed under a pre-registered larger-N protocol.

Strong test. When an LLM receives a CFLT-formatted prompt versus a semantically identical scrambled-order prompt, attention weights to Slot 0 should be higher in the CFLT condition —measurable *descriptively* via attention rollout (Abnar & Zuidema 2020) or attention visualization (Vig 2019), neither of which is a causal method on its own — and downstream task accuracy should be higher for tasks where the Core constituent is the answer-bearing element, across ≥ 3 model families. Establishing that the attention concentration *causes* the accuracy advantage, rather than merely co-occurring with it, requires intervention-based attribution (ablation, activation patching, or causal tracing) on open-weight models, reported separately from the descriptive attention diagnostics.

Falsification condition. If across ≥ 3 frontier model families the CFLT-formatted prompt shows neither attention-weight concentration at position 0 nor task-accuracy advantage, the LLM-side claim is refuted.

Stronger version —protocol vs. alignment constraint. The pilot tests CFLT at prompt-time. A stronger claim is that training-time exposure to CFLT-conformant data would reduce LLM *intrinsic* order-sensitivity. A small-scale fine-tuning experiment —models fine-tuned on CFLT-reordered versus naturally ordered instruction data, evaluated under prompt-order perturbation —would distinguish CFLT-as-prompt-protocol from CFLT-as-alignment-constraint. The latter is the more substantial contribution to AI alignment.

7.4 Neural —Bilingual PFC cost reduction (P3)

Claim. Bilingual speakers trained on a CFLT intermediate scaffold will show reduced prefrontal-cortex (PFC) activation during L1→L2 production tasks compared to bilinguals using ad-hoc surface restructuring (Abutalebi & Green 2007, 2016), as measured by functional MRI (fMRI) or functional near-infrared spectroscopy (fNIRS).

Methodology. Within-subjects design with CFLT-trained vs. untrained translation conditions; measure blood-oxygen-level-dependent (BOLD) signal in two regions that Seeley et al. (2007) show are *dissociable* —the dorsolateral PFC (a node of the executive-control network) and the dorsal anterior cingulate (a node of the salience network) —with a separate hypothesis for each rather than treating them as a single network; draw motivation from the cross-linguistic word-order neuroimaging literature surveyed by Hashimoto, Yokoyama & Kawashima (2012), whose lab also presented preliminary findings on cross-linguistic word-order modulation of brain response as a conference abstract in 2006 (Yokoyama, Fukushima, Riera & Kawashima 2006, *NeuroImage* OHBM abstract).[11]

Falsification condition. If trained CFLT speakers show equal or greater PFC activation than untrained controls, the cognitive-cost-reduction claim is refuted.

7.5 Formal-semantic —CFLT at the syntax–semantics interface (Exploratory)

This subsection identifies a research direction rather than a single falsifiable claim, and is labeled exploratory accordingly.

Can the Tier 1 / Tier 2 partition be characterized formally as a constraint at the syntax-semantics interface —for example, as a principled cut in scope-taking behavior, with Tier 2 adjuncts functioning as world-restricting modifiers?

[11] Hashimoto et al. 2012 is a literature survey of prior SVO/SOV imaging studies, not a single new experiment; it does not supply a directly reusable protocol or test Figure-Ground reversal, so the present study must select a primary experimental paradigm of its own. The 2012 paper appears in *The Open Medical Imaging Journal* (Bentham Open), whose venue has weaker peer-review scrutiny than the canonical neuroimaging journals; it is therefore treated as indirect neurotypological motivation rather than an evidential anchor, and the present study's contrast should replicate against an independent imaging cohort before drawing strong conclusions.

Engagement with the formal-semantic tradition would convert CFLT from an engineering heuristic into a testable claim about the architecture of meaning composition. The natural reference points are Heim & Kratzer (1998) on compositional semantics for the Tier 1 / Tier 2 cut itself; Kratzer (1991, 2012) on modality and conditionals for the Tier 2 Reason / Condition slot as a world-restricting modifier in the sense of restrictor / nuclear-scope partition; and Krifka (1989, 1998) on event composition and the origins of telicity for the Tier 1 event nucleus, its participants, and its aspectual closure. A concrete first-stage falsifiable sub-claim within this program: the Tier 2 adjunct ordering should correspond to scope-nesting in compositional semantics, with Reason scoping over Space scoping over Time, such that swapping the surface order without re-bracketing produces ill-formed or differently-truth-conditioned readings. If a corpus / native-speaker judgment study across the surveyed five-language range shows that scope-nesting is order-invariant for these adjunct classes, the formal-semantic embedding of CFLT collapses to "discourse convention" and the interface-economy reading is refuted.

7.6 Human–AI synchronized reasoning (Exploratory)

This subsection is labeled exploratory because the operationalization of "drift" is not yet pre-committed.

If both human prompt authors and LLM responders are CFLT-conformant, does multi-turn agentic reasoning exhibit lower drift than under unconstrained protocols? Concrete first-stage operationalization: on the SWE-bench (or GAIA) benchmark, agents instructed under a CFLT system prompt should show (a) higher per-turn task-relevant-token ratio, and (b) lower across-turn topic-divergence (measured as cosine distance between consecutive turn embeddings) than agents under a matched unconstrained-protocol baseline, across ≥ 3 frontier model families. Falsification condition: if neither (a) nor (b) improves at $p < .05$ across the model families, the human-AI synchronization claim is refuted at the agent-workflow scale. Stronger versions of the claim —that human-side CFLT training plus LLM-side CFLT prompting compose multiplicatively rather than additively —are deferred until single-side claims are independently confirmed.

A concrete agentic special case —whether CFLT-ordered intent improves first-attempt tool-call success and reduces over-permission incidents when an agent acts through structured tool-call interfaces (e.g., MCP-style function schemas) —is continuous with this test. We flag it as an open question, not a separate contribution, and make no claim that a dedicated translation layer is required beyond existing LLM tool-use scaffolding.

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8. Conclusion

We have argued that linearization cost is the shared currency of two superficially distant problems —adult L2 production and human-to-LLM prompting —and that a single normative discourse-level protocol can pay that cost in both currencies. The protocol fixes the relative order of two tiers (`[Core] → [Reason] → [Space] → [Time]`) while leaving each language's internal morphosyntax intact, and it leaves the contents of each slot to the speaker.

We have been careful to mark several things this paper does *not* claim:

- CFLT is **not** a descriptive universal of natural-language word order; it is a normative protocol that operates one level up from surface syntax.
- CFLT does **not** depend on strong Universal Grammar; the core motivation is Levelt's (1989) preverbal message formation, which is compatible with usage-based, constructionist, and anti-UG positions.
- The **R-S-T inner order is a convention with rationale**, not a derivation; it may be revised under empirical evaluation, particularly for narrative-genre production.
- CFLT exploits **primacy**, not the attention-sink artifact (Xiao et al. 2024); the two mechanisms are distinct and should not be conflated.
- The pilot evidence (§6) is **suggestive, not confirmatory**, and includes one informative null/negative result (Level 4 buried-decision condition on one model) that constrains rather than supports the theory.

CFLT is best read not as a finished theory but as a research scaffold whose value will be determined by the open questions of §7 —across psycholinguistics, typology, LLM mechanistic interpretability, neuroscience, formal semantics, and human-AI interaction. Whether the specific protocol we have proposed is the correct one is, ultimately, an empirical question; what we hope this paper establishes is that the question is worth asking, and that the falsification conditions for asking it well are now on the table.

The empirical agenda enumerated in §7 —six sub-programs (§7.1–§7.6) spanning psycholinguistic production, cross-linguistic corpus work, LLM mechanistic interpretability, bilingual neuroimaging, formal-semantic embedding, and human-AI multi-turn coordination —is intended as a coherent multi-phase open research program. Whether a single Core-then-Frame ordering can additionally carry intent into the structured tool-call interfaces through which LLM agents act is flagged as an open question in §7.6, not a contribution of this paper. The author is independently developing this program at https://cflt.center and welcomes correspondence from researchers whose work intersects any of the open sub-areas.

Data and Code Availability

Prompt templates, raw API responses, the dataset (`dataset.json` v2.0.0; 24 cases × 4 levels × 2 languages), the evaluation scripts (`scripts/llm_eval/`), and the May 2026 report archive (`experiment-history/2026-05-17/`) are deposited at https://cflt.center. The full ablation reproduces in a single command (`python scripts/llm_eval/part2_llm_cflt_eval.` `--runs 3`). A pre-registered protocol for the confirmatory study described in §7 will be filed at a public pre-registration registry prior to data collection.

A reference implementation (human bilingual education) is independently maintained as an open-source application (Next.js + Electron) at https://github.com/corefirst/corefirst (Apache 2.0). It runs fully locally given a valid LLM API key. The pilot empirical evaluation in §6 is independent of this implementation and relies only on the standalone scripts in `scripts/llm_eval/`; the implementation is referenced here as concrete evidence that the protocol is operationally deployable, not as part of §6's empirical evidence base.

Conflict of Interest

The author maintains the cflt.center project repository, which serves as the open materials site for this work and as the canonical specification of the CFLT framework. The framework itself is licensed CC BY 4.0 and is not a commercial product. The author declares no financial conflicts of interest.

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Ethics

This study involved no human or animal subjects. The pilot empirical evaluation (§6) consisted entirely of commercial Large Language Model API queries against publicly accessible model endpoints; no personal data, user-generated content, or identifiable information was processed. The bilingual test items in `scripts/llm_eval/dataset.json` were authored by the project lead for benchmark purposes and contain no personal or sensitive content. No ethics board approval was sought, as no human-subjects research was conducted; this paper reports a purely computational ablation.

AI Use Disclosure

This paper's AI-assisted workflow is disclosed in accordance with the 2025–2026 standards adopted by major academic venues.

The author, a non-native English speaker, conducted all core theoretical development, experimental design, and data analysis independently. AI tools (Claude, Gemini) were employed exclusively for grammatical copy-editing, stylistic refinement, and academic register optimization of the author's original English drafts. All output was rigorously reviewed and revised by the human author to ensure conceptual accuracy. The author maintains full accountability for the entirety of this work.

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This paper summarizes work developed within the CFLT framework (cflt.center). The framework integrates research across cognitive linguistics, formal semantics, psycholinguistics, second-language acquisition, neuroscience, and NLP; the full canonical specification, with bibliography of approximately 150 entries, is maintained at the project site. The author thanks the maintainers of the five frontier LLM platforms evaluated in this work. Any errors of theoretical framing or empirical interpretation remain the author's own.

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A complete bibliography of approximately 150 entries spanning all eight foundational disciplines is maintained at https://cflt.center/bibliography.